

ASSIGNMENT 2

QUESTION 1

Write a program to find the complement of an image.

SOLUTION

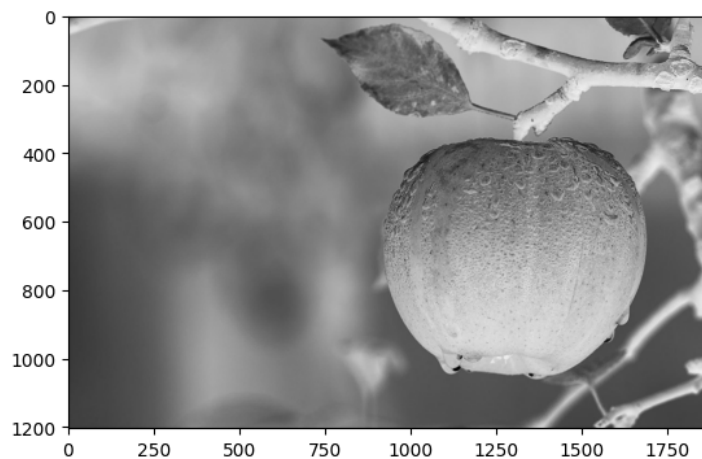
```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("./gray.jpeg")
# print(img)

complement = 255 - img
plt.imshow(complement)
```

OUTPUT

<matplotlib.image.AxesImage at 0x7f58d19bead0><Figure size 640x480 with 1 Axes>



QUESTION 2

Convert image to grayscale using

- (a) Mean threshold
- (b) User-defined threshold

SOLUTION

```
import cv2
import numpy as np

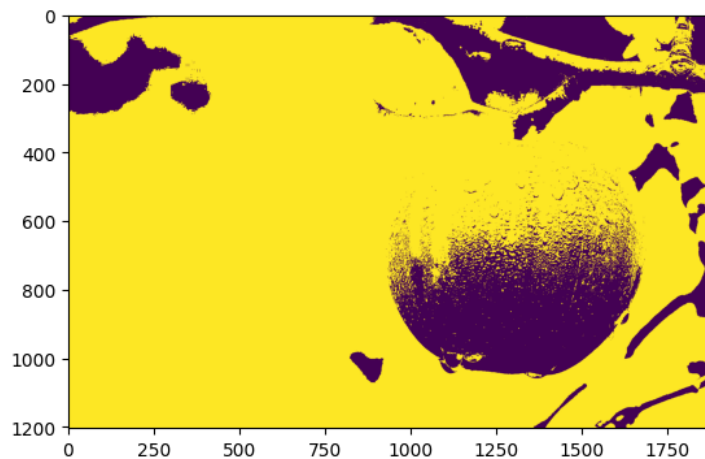
img = cv2.imread('./gray.jpeg', cv2.IMREAD_GRAYSCALE)

# Using mean threshold
_, binary_img = cv2.threshold(img, np.mean(img), 255, cv2.THRESH_BINARY)
plt.imshow(binary_img)
```

```
# Usin user-defined threshold
threshold = 80
_, binary_img = cv2.threshold(img, threshold, 255, cv2.THRESH_BINARY)
plt.imshow(binary_img)
```

OUTPUT

<matplotlib.image.AxesImage at 0x7f58d17dd810><Figure size 640x480 with 1 Axes>



QUESTION 3

Convert a colored image to a grayscale image by:
(a) taking mean average of all 3 planes
(b) User-defined weightage

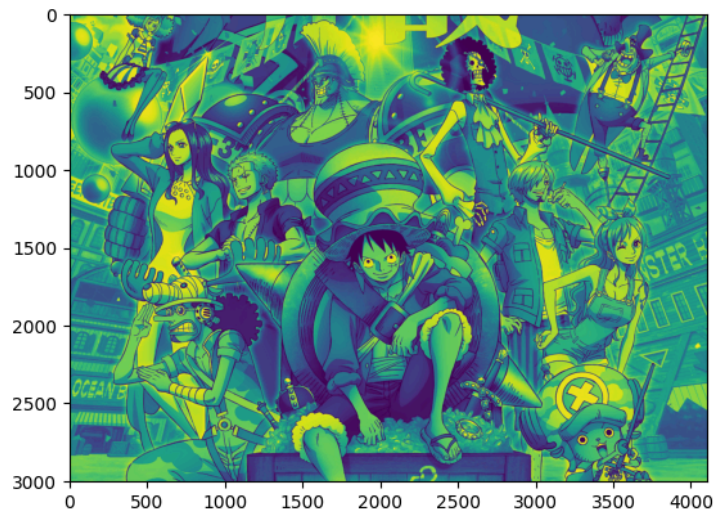
SOLUTION

```
img = cv2.imread('./color.jpg')

grayscale = np.mean(img, axis=2)
# convert grayscale to unsigned integers (get rid of decimals/floating point numbers)
grayscale.astype(np.uint8)
plt.imshow(grayscale)
```

OUTPUT

<matplotlib.image.AxesImage at 0x7f58d1a94b90><Figure size 640x480 with 1 Axes>



SOLUTION

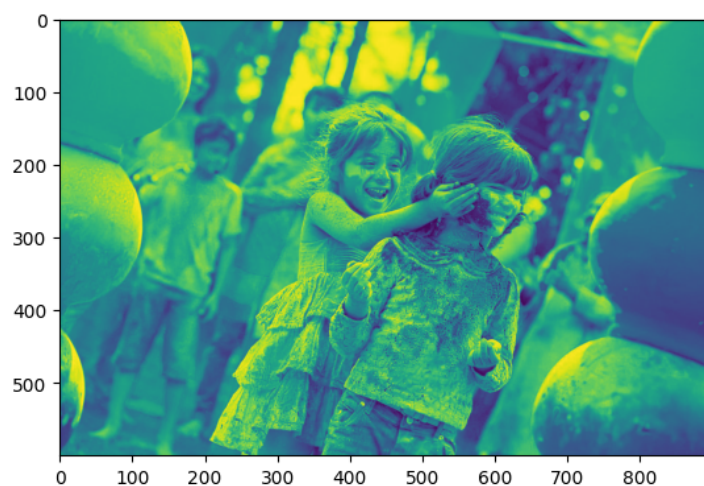
```
weightage_r = 0.6
weightage_g = 0.0
weightage_b = 0.4

img = cv2.imread('./holi.jpg')
gray_img = (weightage_r * img[:, :, 2]) + (weightage_g * img[:, :, 1]) + (weightage_b *
img[:, :, 0])

plt.imshow(gray_img)
```

OUTPUT

<matplotlib.image.AxesImage at 0x7f58d1054310><Figure size 640x480 with 1 Axes>



QUESTION 4

Draw a border around an image.

SOLUTION

```
img = cv2.imread('./gray.jpeg', cv2.IMREAD_GRAYSCALE)
border_width = 200

img_with_border = cv2.copyMakeBorder(img, border_width, border_width, border_width,
border_width, cv2.BORDER_CONSTANT, value=(255, 0, 0))
plt.imshow(img_with_border)
```

OUTPUT

<matplotlib.image.AxesImage at 0x7f58d0edfbd0><Figure size 640x480 with 1 Axes>

